

Practice Problem Set

Introduction to Statistical Methods

For My Sec9 Students

August 26, 2025

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1 Descriptive Statistics and Probability

This section covers foundational concepts including measures of central tendency, variability, data distribution shapes, and the fundamentals of probability theory.

Problem 1.1. The following data represents the time (in minutes) taken by 10 employees to complete a task: 39, 29, 43, 52, 39, 44, 40, 31, 44, 35.

- (a) Calculate the mean, median, and mode.
- (b) Calculate the range, variance, and standard deviation.
- (c) Based on the mean and median, comment on the skewness of the data.

Solution. (a) Measures of Central Tendency

Step 1: Calculate the Mean. The mean is the sum of the values divided by the count.

$$\bar{x} = \frac{39 + 29 + 43 + 52 + 39 + 44 + 40 + 31 + 44 + 35}{10} = \frac{396}{10} = 39.6$$

Step 2: Calculate the Median. First, sort the data in ascending order: 29, 31, 35, 39, 39, 40, 43, 44, 44, 52. Since there is an even number of observations (n=10), the median is the average of the two middle values (5th and 6th).

$$\text{Median} = \frac{39 + 40}{2} = 39.5$$

Step 3: Identify the Mode. The mode is the most frequently occurring value. The values 39 and 44 both appear twice, which is more than any other value. Thus, the data is bimodal.

$$\text{Mode} = 39 \text{ and } 44$$

(b) Measures of Variability

Step 1: Calculate the Range. Range = Maximum Value - Minimum Value.

$$\text{Range} = 52 - 29 = 23$$

Step 2: Calculate the Sample Variance (s^2). The formula for sample variance is $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$.

$$\begin{aligned}\sum(x_i - \bar{x})^2 &= (29 - 39.6)^2 + (31 - 39.6)^2 + \dots + (52 - 39.6)^2 \\ &= (-10.6)^2 + (-8.6)^2 + (-4.6)^2 + (-0.6)^2 + (-0.6)^2 \\ &\quad + (0.4)^2 + (3.4)^2 + (4.4)^2 + (4.4)^2 + (12.4)^2 \\ &= 112.36 + 73.96 + 21.16 + 0.36 + 0.36 + 0.16 + 11.56 + 19.36 + 19.36 + 153.76 \\ &= 412.4\end{aligned}$$

$$s^2 = \frac{412.4}{10 - 1} = \frac{412.4}{9} \approx 45.82$$

Step 3: Calculate the Standard Deviation (s). The standard deviation is the square root of the variance.

$$s = \sqrt{45.82} \approx 6.77$$

(c) Comment on Skewness

Step 1: Compare Mean and Median. We calculated Mean = 39.6 and Median = 39.5.

Step 2: Conclusion. Since the mean (39.6) is slightly greater than the median (39.5), the data is slightly **positively skewed** (or right-skewed). This indicates a slight tail towards the higher values.

Problem 1.2. In a university, 60% of students have a laptop. Among students with a laptop, 75% also have a tablet. 45% of all students have a tablet.

- (a) What is the probability that a randomly selected student has both a laptop and a tablet?
- (b) What is the probability that a student has a laptop or a tablet (or both)?
- (c) Are the events "has a laptop" and "has a tablet" independent? Justify your answer.

Solution. Let L be the event that a student has a laptop, and T be the event that a student has a tablet. We are given: $P(L) = 0.60$, $P(T) = 0.45$, and the conditional probability $P(T|L) = 0.75$.

(a) Probability of having both a laptop and a tablet

Step 1: Use the definition of conditional probability. The formula is $P(T \cap L) = P(T|L) \times P(L)$.

Step 2: Calculate the probability.

$$P(T \cap L) = 0.75 \times 0.60 = 0.45$$

The probability that a student has both is 45%.

(b) Probability of having a laptop or a tablet

Step 1: Use the general addition rule for probability. The formula is $P(L \cup T) = P(L) + P(T) - P(L \cap T)$.

Step 2: Calculate the probability.

$$P(L \cup T) = 0.60 + 0.45 - 0.45 = 0.60$$

The probability that a student has either a laptop or a tablet is 60%.

(c) Independence of Events

Step 1: State the condition for independence. Two events L and T are independent if and only if $P(L \cap T) = P(L) \times P(T)$.

Step 2: Check the condition.

- From part (a), we know $P(L \cap T) = 0.45$.
- Calculate the product of individual probabilities: $P(L) \times P(T) = 0.60 \times 0.45 = 0.27$.

Step 3: Conclusion. Since $0.45 \neq 0.27$, the events "has a laptop" and "has a tablet" are **not independent**.

Problem 1.3. A systems manager tracks the number of server failures per day. For the past two weeks, the data is: 1, 3, 0, 3, 2, 3, 4, 3, 0, 1, 3, 2, 6, 4. Construct a five-number summary for this data and identify any potential outliers using the $1.5 \times \text{IQR}$ rule.

Solution.

Step 1: Sort the data. First, arrange the 14 data points in ascending order:

$$0, 0, 1, 1, 2, 2, 3, 3, 3, 3, 3, 4, 4, 6$$

Step 2: Find the Minimum and Maximum.

- Minimum = 0
- Maximum = 6

Step 3: Find the Median (Q2). With $n = 14$ (an even number), the median is the average of the 7th and 8th values.

$$\text{Median (Q2)} = \frac{3 + 3}{2} = 3$$

Step 4: Find the First Quartile (Q1). Q1 is the median of the lower half of the data: $\{0, 0, 1, 1, 2, 2, 3\}$. The middle value is the 4th value.

$$Q1 = 1$$

Step 5: Find the Third Quartile (Q3). Q3 is the median of the upper half of the data: $\{3, 3, 3, 3, 4, 4, 6\}$. The middle value is the 4th value of this set.

$$Q3 = 3$$

Step 6: State the Five-Number Summary. The five-number summary is: **Minimum = 0, Q1 = 1, Median = 3, Q3 = 3, Maximum = 6.**

Step 7: Identify Potential Outliers.

- Calculate the Interquartile Range (IQR): $IQR = Q3 - Q1 = 3 - 1 = 2$.
- Calculate the lower fence: $Q1 - 1.5 \times IQR = 1 - 1.5 \times 2 = 1 - 3 = -2$.
- Calculate the upper fence: $Q3 + 1.5 \times IQR = 3 + 1.5 \times 2 = 3 + 3 = 6$.

Step 8: Conclusion. Any data point below -2 or above 6 is a potential outlier. In our dataset, the minimum is 0 (which is greater than -2) and the maximum is 6 (which is not greater than the upper fence of 6). Therefore, according to the $1.5 \times \text{IQR}$ rule, there are **no outliers** in this dataset.

Problem 1.4. At a factory, three machines (A, B, and C) produce bolts. Machine A produces 40% of the bolts, Machine B produces 35%, and Machine C produces 25%. The defective rates for these machines are 5%, 3%, and 2%, respectively. A bolt is selected at random from the total output and is found to be defective. What is the probability that it was produced by Machine A?

Solution. This is a classic application of Bayes' Theorem. Let A, B, C be the events that a bolt is produced by Machine A, B, C respectively. Let D be the event that a bolt is defective.

Step 1: List the known probabilities.

- Priors: $P(A) = 0.40$, $P(B) = 0.35$, $P(C) = 0.25$.
- Likelihoods (Conditional Probabilities): $P(D|A) = 0.05$, $P(D|B) = 0.03$, $P(D|C) = 0.02$.

Step 2: State the Goal. We want to find the posterior probability $P(A|D)$.

Step 3: Apply Bayes' Theorem Formula.

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)}$$

Step 4: Calculate the Total Probability of a Defect, $P(D)$. We use the Law of Total Probability:

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C) \\ &= (0.05)(0.40) + (0.03)(0.35) + (0.02)(0.25) \\ &= 0.0200 + 0.0105 + 0.0050 \\ &= 0.0355 \end{aligned}$$

The overall probability of picking a defective bolt is 3.55%.

Step 5: Calculate the Final Posterior Probability.

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{(0.05)(0.40)}{0.0355} = \frac{0.0200}{0.0355} \approx 0.5634$$

Step 6: Conclusion. Given that a defective bolt is selected, the probability that it came from Machine A is approximately **56.34%**.

Problem 1.5. Two dice are thrown. Let A be the event that the sum is greater than 8, and B be the event that at least one die shows a 6.

- (a) Find $P(A)$.
- (b) Find $P(B)$.
- (c) Find the conditional probability $P(A|B)$.

Solution. The sample space consists of 36 equally likely outcomes.

(a) Find $P(A)$

Step 1: List the outcomes for event A (sum ≥ 8). The possible sums are 9, 10, 11, 12.

- Sum=9: (3,6), (4,5), (5,4), (6,3)
- Sum=10: (4,6), (5,5), (6,4)
- Sum=11: (5,6), (6,5)
- Sum=12: (6,6)

There are $4+3+2+1 = 10$ favorable outcomes.

Step 2: Calculate the probability.

$$P(A) = \frac{\text{Number of outcomes in A}}{\text{Total outcomes}} = \frac{10}{36} = \frac{5}{18}$$

(b) Find P(B)

Step 1: List the outcomes for event B (at least one 6).

- Die 1 is 6: (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)
- Die 2 is 6 (excluding (6,6) which is already counted): (1,6), (2,6), (3,6), (4,6), (5,6)

There are $6+5 = 11$ favorable outcomes.

Step 2: Calculate the probability.

$$P(B) = \frac{11}{36}$$

(c) Find P(A—B)

Step 1: Use the formula for conditional probability.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Step 2: Find the intersection $A \cap B$. This is the event that the sum is greater than 8 AND at least one die is a 6. We look at the outcomes for A and see which ones have a 6. Outcomes from A: (3,6), (4,5), (5,4), (6,3), (4,6), (5,5), (6,4), (5,6), (6,5), (6,6). Outcomes with a 6: (3,6), (6,3), (4,6), (6,4), (5,6), (6,5), (6,6). There are 7 outcomes in the intersection. So, $P(A \cap B) = \frac{7}{36}$.

Step 3: Calculate the conditional probability.

$$P(A|B) = \frac{7/36}{11/36} = \frac{7}{11}$$

Step 4: Conclusion. The probability that the sum is greater than 8, given that at least one die shows a 6, is **7/11**.

Problem 1.6. A committee of 5 is to be chosen from a group of 8 men and 4 women. What is the probability that the committee contains a majority of women?

Solution.

Step 1: Define the total number of possible committees. There are a total of $8 + 4 = 12$ people. The total number of ways to choose a committee of 5 is given by the combination formula:

$$\text{Total Committees} = \binom{12}{5} = \frac{12!}{5!(12-5)!} = \frac{12 \times 11 \times 10 \times 9 \times 8}{5 \times 4 \times 3 \times 2 \times 1} = 792$$

Step 2: Define the favorable outcomes. A majority of women means the committee must have more women than men. Since the committee size is 5, this means having 3, 4, or 5 women. As there are only 4 women available, the possibilities are:

- Case 1: 3 women and 2 men.
- Case 2: 4 women and 1 man.

Step 3: Calculate the number of ways for each case.

- Case 1: $\binom{4}{3} \times \binom{8}{2} = 4 \times \frac{8 \times 7}{2} = 4 \times 28 = 112$
- Case 2: $\binom{4}{4} \times \binom{8}{1} = 1 \times 8 = 8$

Step 4: Calculate the total number of favorable outcomes. Total favorable outcomes = $112 + 8 = 120$.

Step 5: Calculate the final probability.

$$P(\text{Majority Women}) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}} = \frac{120}{792} = \frac{10}{66} = \frac{5}{33}$$

Problem 1.7. For two events A and B, we have $P(A \cup B) = 0.8$, $P(A) = 0.5$, and $P(B|A) = 0.6$. Find $P(B)$.

Solution.

Step 1: Find the intersection probability, $P(A \cap B)$. Using the conditional probability formula:

$$P(A \cap B) = P(B|A) \times P(A) = 0.6 \times 0.5 = 0.30$$

Step 2: Use the general addition rule to find $P(B)$. The rule is $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. We can rearrange this to solve for $P(B)$.

$$P(B) = P(A \cup B) - P(A) + P(A \cap B)$$

Step 3: Substitute the known values.

$$P(B) = 0.8 - 0.5 + 0.3 = 0.6$$

Step 4: Conclusion. The probability of event B is **0.6**.

Problem 1.8. You are presented with two datasets of salaries for a department.

- Dataset 1: \$50k, \$52k, \$55k, \$58k, \$60k
- Dataset 2: \$50k, \$52k, \$55k, \$58k, \$150k

For each dataset, calculate the mean and median.

Solution.

Step 1: Calculate Mean and Median for Dataset 1.

- Mean: $\frac{50+52+55+58+60}{5} = \frac{275}{5} = \$55k$
- Median: The data is sorted. The middle value (3rd) is \$55k.

Step 2: Calculate Mean and Median for Dataset 2.

- Mean: $\frac{50+52+55+58+150}{5} = \frac{365}{5} = \$73k$
- Median: The data is sorted. The middle value (3rd) is \$55k.

Step 3: Conclusion. For Dataset 1, Mean = \$55k and Median = \$55k. For Dataset 2, Mean = \$73k and Median = \$55k. The median is a more appropriate measure of central tendency for Dataset 2 because it is robust to the outlier (\$150k).

Problem 1.9. Let the sample space be $S = \{1, 2, 3, 4, 5, 6\}$. Let A be the event $\{1, 2, 3\}$ and B be the event $\{3, 4, 5\}$. Find the probabilities of the following events:

- (a) $P(A \text{ and } B)$, i.e., $P(A \cap B)$
- (b) $P(A \text{ or } B)$, i.e., $P(A \cup B)$
- (c) $P(A^c)$, the complement of A

Solution. Assuming each outcome in the sample space is equally likely, the probability of any single outcome is $1/6$.

(a) Probability of A and B

Step 1: Find the intersection of A and B. $A \cap B = \{1, 2, 3\} \cap \{3, 4, 5\} = \{3\}$.

Step 2: Calculate the probability. The intersection contains one outcome.

$$P(A \cap B) = \frac{1}{6}$$

(b) Probability of A or B

Step 1: Find the union of A and B. $A \cup B = \{1, 2, 3\} \cup \{3, 4, 5\} = \{1, 2, 3, 4, 5\}$.

Step 2: Calculate the probability. The union contains five outcomes.

$$P(A \cup B) = \frac{5}{6}$$

(c) Probability of the complement of A

Step 1: Find the complement of A. $A^c = S - A = \{1, 2, 3, 4, 5, 6\} - \{1, 2, 3\} = \{4, 5, 6\}$.

Step 2: Calculate the probability. The complement contains three outcomes.

$$P(A^c) = \frac{3}{6} = \frac{1}{2}$$

Problem 1.10. A survey of a city's population showed that 40% read Newspaper A, 25% read Newspaper B, and 15% read both.

- (a) What percentage of the population reads at least one of the newspapers?
- (b) What percentage of the population reads exactly one newspaper?
- (c) Of those who read Newspaper A, what percentage also read Newspaper B?

Solution. Let A be the event of reading Newspaper A and B be the event of reading Newspaper B. Given: $P(A) = 0.40$, $P(B) = 0.25$, $P(A \cap B) = 0.15$.

(a) Percentage reading at least one newspaper

Step 1: Identify the required probability. "At least one" corresponds to the union of the two events, $P(A \cup B)$.

Step 2: Apply the addition rule.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.40 + 0.25 - 0.15 = 0.50$$

Step 3: Conclusion. 50% of the population reads at least one newspaper.

(b) Percentage reading exactly one newspaper

Step 1: Formulate the probability. This corresponds to the outcomes in the union minus the outcomes in the intersection.

$$P(\text{Exactly One}) = P(A \cup B) - P(A \cap B)$$

Step 2: Calculate the value.

$$P(\text{Exactly One}) = 0.50 - 0.15 = 0.35$$

Step 3: Conclusion. 35% of the population reads exactly one newspaper.

(c) Percentage who read B, given they read A

Step 1: Identify the required probability. This is the conditional probability $P(B|A)$.

Step 2: Apply the conditional probability formula.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.15}{0.40} = \frac{15}{40} = \frac{3}{8} = 0.375$$

Step 3: Conclusion. Of those who read Newspaper A, 37.5% also read Newspaper B.

2 Probability Distributions

This section explores various discrete and continuous probability distributions, including their properties, applications, and the calculation of key metrics like mean and variance.

Problem 2.1. A call center receives an average of 4 calls per minute. Assuming the calls follow a Poisson distribution, find the probability that in a given minute:

- (a) Exactly 2 calls are received.
- (b) At least 3 calls are received.
- (c) No calls are received in a 30-second interval.

Solution. The Poisson probability mass function is given by $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, where λ is the average rate of events.

(a) Exactly 2 calls are received

Step 1: Identify parameters. For a one-minute interval, $\lambda = 4$. We want to find $P(X = 2)$.

Step 2: Apply the formula.

$$P(X = 2) = \frac{e^{-4} 4^2}{2!} = \frac{e^{-4} \cdot 16}{2} = 8e^{-4} \approx 0.1465$$

(b) At least 3 calls are received

Step 1: Formulate the probability. "At least 3" means $P(X \geq 3)$. It's easier to calculate the complement: $1 - P(X < 3)$.

$$P(X \geq 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

Step 2: Calculate the individual probabilities. We have $\lambda = 4$.

- $P(X = 0) = \frac{e^{-4} 4^0}{0!} = e^{-4} \approx 0.0183$
- $P(X = 1) = \frac{e^{-4} 4^1}{1!} = 4e^{-4} \approx 0.0733$
- $P(X = 2) \approx 0.1465$ (from part a)

Step 3: Calculate the final probability.

$$P(X \geq 3) = 1 - (0.0183 + 0.0733 + 0.1465) = 1 - 0.2381 = 0.7619$$

(c) No calls in a 30-second interval

Step 1: Adjust the rate λ for the new interval. The original rate is 4 calls per 60 seconds. For a 30-second interval, the new average rate is:

$$\lambda' = 4 \times \frac{30}{60} = 2$$

Step 2: Calculate the probability of zero events. We want to find $P(X = 0)$ with $\lambda' = 2$.

$$P(X = 0) = \frac{e^{-2} 2^0}{0!} = e^{-2} \approx 0.1353$$

Problem 2.2. A machine produces items with a 10% defect rate. A quality inspector randomly selects 8 items. Let X be the number of defective items found.

- (a) What is the probability that exactly 2 items are defective?
- (b) What is the probability that at most 1 item is defective?
- (c) What are the mean and variance of the number of defective items?

Solution. This scenario follows a Binomial distribution. The probability mass function is $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.

(a) Probability of exactly 2 defects

Step 1: Identify parameters. $n = 8, p = 0.1, k = 2$.

Step 2: Apply the formula.

$$P(X = 2) = \binom{8}{2} (0.1)^2 (0.9)^{8-2} = \frac{8!}{2!6!} (0.01)(0.9)^6 = 28 \times 0.01 \times 0.531441 \approx 0.1488$$

(b) Probability of at most 1 defect

Step 1: Formulate the probability. "At most 1" means $P(X \leq 1) = P(X = 0) + P(X = 1)$.

Step 2: Calculate the individual probabilities.

- $P(X = 0) = \binom{8}{0} (0.1)^0 (0.9)^8 = 1 \times 1 \times 0.430467 \approx 0.4305$
- $P(X = 1) = \binom{8}{1} (0.1)^1 (0.9)^7 = 8 \times 0.1 \times 0.478297 \approx 0.3826$

Step 3: Calculate the final probability.

$$P(X \leq 1) = 0.4305 + 0.3826 = 0.8131$$

(c) Mean and Variance

Step 1: Use the formulas for the Binomial distribution. Mean (μ) = np ; Variance (σ^2) = $np(1 - p)$.

Step 2: Calculate the values.

- $\mu = 8 \times 0.1 = 0.8$
- $\sigma^2 = 8 \times 0.1 \times 0.9 = 0.72$

Problem 2.3. The height of adult males in a city is normally distributed with a mean of 175 cm and a standard deviation of 7 cm.

- (a) What is the probability that a randomly selected male is taller than 185 cm?
- (b) What is the probability that a randomly selected male has a height between 170 cm and 180 cm?
- (c) What height is the 90th percentile?

Solution. We use the standard normal distribution by converting the height X to a Z-score using the formula $Z = \frac{X - \mu}{\sigma}$. Here, $\mu = 175$ and $\sigma = 7$.

(a) Probability of being taller than 185 cm

Step 1: Calculate the Z-score for $X = 185$.

$$Z = \frac{185 - 175}{7} = \frac{10}{7} \approx 1.43$$

Step 2: Find the probability from the Z-table. We want $P(Z > 1.43) = 1 - P(Z < 1.43)$.

$$P(Z > 1.43) = 1 - 0.9236 = 0.0764$$

(b) Probability of height between 170 and 180 cm

Step 1: Calculate the Z-scores for $X=170$ and $X=180$.

$$Z_1 = \frac{170 - 175}{7} \approx -0.71 \quad , \quad Z_2 = \frac{180 - 175}{7} \approx 0.71$$

Step 2: Find the probability from the Z-table. We want $P(-0.71 < Z < 0.71) = P(Z < 0.71) - P(Z < -0.71)$.

$$P(-0.71 < Z < 0.71) = 0.7611 - 0.2389 = 0.5222$$

(c) Height at the 90th percentile

Step 1: Find the Z-score corresponding to the 90th percentile. We look for the Z-score such that $P(Z < z) = 0.90$. From the Z-table, the closest Z-score is approximately 1.28.

Step 2: Convert the Z-score back to the original scale (X). Use $X = \mu + Z\sigma$.

$$X = 175 + (1.28)(7) = 175 + 8.96 = 183.96$$

The 90th percentile for height is approximately 183.96 cm.

Problem 2.4. Let X and Y be two discrete random variables with the following joint probability mass function $f(x, y)$:

$f(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	0.1	0.2	0.1
$x=1$	0.3	0.1	0.2

- (a) Find the marginal probability distributions of X and Y.
- (b) Calculate the expected value of X, $E(X)$, and the expected value of Y, $E(Y)$.
- (c) Are X and Y independent? Justify your answer.

Solution. (a) Marginal Distributions

Step 1: Find the marginal distribution of X, $g(x)$. Sum probabilities across rows.

- $g(0) = P(X = 0) = 0.1 + 0.2 + 0.1 = 0.4$
- $g(1) = P(X = 1) = 0.3 + 0.1 + 0.2 = 0.6$

Step 2: Find the marginal distribution of Y, $h(y)$. Sum probabilities down columns.

- $h(0) = P(Y = 0) = 0.1 + 0.3 = 0.4$

- $h(1) = P(Y = 1) = 0.2 + 0.1 = 0.3$
- $h(2) = P(Y = 2) = 0.1 + 0.2 = 0.3$

(b) Expected Values

Step 1: Calculate $E(X)$. $E(X) = \sum x \cdot g(x) = (0 \times 0.4) + (1 \times 0.6) = 0.6$.

Step 2: Calculate $E(Y)$. $E(Y) = \sum y \cdot h(y) = (0 \times 0.4) + (1 \times 0.3) + (2 \times 0.3) = 0 + 0.3 + 0.6 = 0.9$.

(c) Independence of X and Y

Step 1: State the condition for independence. X and Y are independent if $f(x, y) = g(x) \times h(y)$ for all (x,y).

Step 2: Test a point, e.g., (x=0, y=0).

- Joint probability: $f(0, 0) = 0.1$.
- Product of marginals: $g(0) \times h(0) = 0.4 \times 0.4 = 0.16$.

Step 3: Conclusion. Since $0.1 \neq 0.16$, the random variables X and Y are **not independent**.

Problem 2.5. The time (in hours) a user spends on a social media app is a continuous random variable X with the probability density function (PDF): $f(x) = \frac{1}{4}x$ for $0 \leq x \leq \sqrt{8}$, and $f(x) = 0$ otherwise.

- Verify that this is a valid PDF.
- Find the probability that a user spends less than 1 hour on the app, $P(X < 1)$.
- Find the expected time spent on the app, $E(X)$.

Solution. (a) Verify the PDF

Step 1: Check for non-negativity. For $0 \leq x \leq \sqrt{8}$, $f(x) = \frac{1}{4}x \geq 0$. The condition is met.

Step 2: Check that the total area under the curve is 1.

$$\int_0^{\sqrt{8}} \frac{1}{4}x \, dx = \frac{1}{4} \left[\frac{x^2}{2} \right]_0^{\sqrt{8}} = \frac{1}{8} [(\sqrt{8})^2 - 0^2] = \frac{8}{8} = 1$$

Since both conditions are met, it is a valid PDF.

(b) Probability of spending less than 1 hour

Step 1: Evaluate the integral.

$$P(X < 1) = \int_0^1 \frac{1}{4}x \, dx = \frac{1}{4} \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{8} (1^2 - 0^2) = \frac{1}{8} = 0.125$$

(c) Expected time spent

Step 1: Evaluate the integral for $E(X)$. $E(X) = \int_0^{\sqrt{8}} x \cdot f(x) \, dx$.

$$E(X) = \int_0^{\sqrt{8}} \frac{1}{4}x^2 \, dx = \frac{1}{4} \left[\frac{x^3}{3} \right]_0^{\sqrt{8}} = \frac{1}{12} [(\sqrt{8})^3] = \frac{16\sqrt{2}}{12} = \frac{4\sqrt{2}}{3} \approx 1.886$$

Problem 2.6. A factory produces light bulbs, and 2% of them are defective. A large shipment of 500 bulbs is sent to a retailer.

- (a) Write the formula for the exact probability that exactly 10 bulbs are defective.
- (b) Use the Poisson approximation to the Binomial distribution to estimate this probability.

Solution. (a) Exact Binomial Probability The exact probability is given by the Binomial formula with $n = 500, p = 0.02, k = 10$:

$$P(X = 10) = \binom{500}{10} (0.02)^{10} (0.98)^{490}$$

(b) Poisson Approximation The Poisson approximation is appropriate since $n = 500$ is large and $p = 0.02$ is small.

Step 1: Calculate the rate parameter λ .

$$\lambda = np = 500 \times 0.02 = 10$$

Step 2: Apply the Poisson formula.

$$P(X = 10) \approx \frac{e^{-\lambda} \lambda^{10}}{10!} = \frac{e^{-10} 10^{10}}{10!} \approx 0.1251$$

Problem 2.7. A continuous random variable X has the cumulative distribution function (CDF):

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x^3/8 & \text{if } 0 \leq x \leq 2 \\ 1 & \text{if } x > 2 \end{cases}$$

- (a) Find the probability density function (PDF), $f(x)$.
- (b) Calculate $P(X > 1)$.
- (c) Find the median of this distribution.

Solution. (a) Find the PDF The PDF is the derivative of the CDF, $f(x) = F'(x)$.

$$f(x) = \frac{d}{dx} \left(\frac{x^3}{8} \right) = \frac{3x^2}{8} \quad \text{for } 0 \leq x \leq 2$$

$$f(x) = \begin{cases} \frac{3x^2}{8} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(b) Calculate $P(X > 1)$ Using the CDF, $P(X > 1) = 1 - P(X \leq 1) = 1 - F(1)$.

$$P(X > 1) = 1 - \frac{1^3}{8} = 1 - \frac{1}{8} = \frac{7}{8}$$

(c) Find the Median The median (m) is the value where the CDF is 0.5.

$$F(m) = 0.5 \implies \frac{m^3}{8} = 0.5 \implies m^3 = 4 \implies m = \sqrt[3]{4} \approx 1.587$$

Problem 2.8. Let X be a random variable with mean $E(X) = 10$ and variance $Var(X) = 4$. Let Y be a new random variable defined as $Y = 3X - 5$. Find the mean and variance of Y .

Solution.

Step 1: Calculate $E(Y)$. Using the property $E(aX + b) = aE(X) + b$:

$$E(Y) = E(3X - 5) = 3E(X) - 5 = 3(10) - 5 = 25$$

Step 2: Calculate $Var(Y)$. Using the property $Var(aX + b) = a^2Var(X)$:

$$Var(Y) = Var(3X - 5) = 3^2Var(X) = 9 \times 4 = 36$$

Step 3: Conclusion. The mean of Y is **25** and the variance of Y is **36**.

Problem 2.9. For the joint PDF $f(x, y) = \frac{x+y}{3}$ for $0 < x < 2, 0 < y < 1$, find the conditional PDF of X given $Y = y$, denoted $f(x|y)$, and calculate $P(X > 1|Y = 0.5)$.

Solution.

Step 1: Find the marginal PDF of Y , $h(y)$.

$$h(y) = \int_0^2 \frac{x+y}{3} dx = \frac{1}{3} \left[\frac{x^2}{2} + yx \right]_0^2 = \frac{1}{3}(2 + 2y) = \frac{2(1+y)}{3}$$

Step 2: Find the conditional PDF $f(x|y)$.

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{(x+y)/3}{2(1+y)/3} = \frac{x+y}{2(1+y)}$$

Step 3: Calculate $P(X > 1|Y = 0.5)$. First, find the specific conditional PDF for $y = 0.5$: $f(x|Y = 0.5) = \frac{x+0.5}{3}$. Now integrate this PDF from 1 to 2.

$$\begin{aligned} P(X > 1|Y = 0.5) &= \int_1^2 \frac{x+0.5}{3} dx = \frac{1}{3} \left[\frac{x^2}{2} + 0.5x \right]_1^2 \\ &= \frac{1}{3} [(2+1) - (0.5+0.5)] = \frac{1}{3}[3-1] = \frac{2}{3} \end{aligned}$$